

Whole grains have been shown to reduce the risk of cardiovascular disorders. A team of scientists discovered that the compound 5-aminovaleric acid betaine (5-AVAB) was present at high levels in cardiomyocytes of mice given a diet rich in whole grains and decided to investigate its role in metabolism. Previous studies showed that 5-AVAB did not affect glycolysis or citric acid cycle rates in cardiomyocytes when glucose was the primary fuel source, but that 5-AVAB did alter fatty acid metabolism.

The plasma membrane transport protein organic cation/carnitine transporter 2 (OCTN2) transports extracellular L-carnitine into the cell against its concentration gradient using the energy provided by the simultaneous transport of Na^+ into the cell. Cellular L-carnitine is required for the transport of fatty acids from the cytosol into the mitochondria. Based on the structures of 5-AVAB and L-carnitine (Figure 1), scientists hypothesized that 5-AVAB may interfere with OCTN2-mediated transport of L-carnitine.

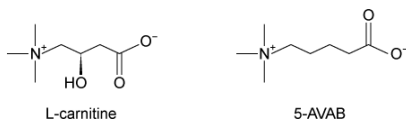


Figure 1 Structures of L-carnitine and 5-AVAB

To test their hypothesis, scientists incubated mouse cardiomyocytes with various concentrations of 5-AVAB for 24 hours. Cells were then provided with the fatty acid palmitate for energy and were deprived of all other energy sources. The cardiomyocytes were also given 50 μM L-carnitine (endogenous extracellular concentration) or 2 mM L-carnitine, and the rate of oxygen consumption was monitored (Figure 2).

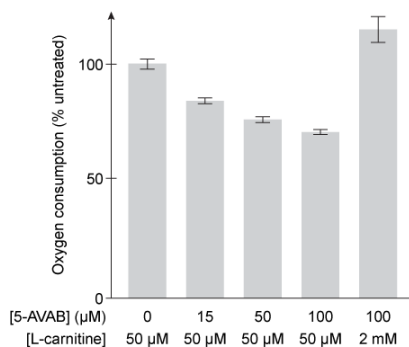


Figure 2 Oxygen consumption rate of cardiomyocytes at various concentrations of 5-AVAB and L-carnitine

In a follow-up experiment, the scientists used siRNA to silence expression of OCTN2 and found that intracellular L-carnitine levels were similarly depleted in siRNA-treated cells as in cells treated with 100 μM 5-AVAB.

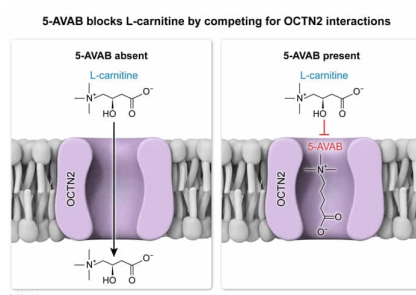
5-AVAB most likely alters fatty acid metabolism in cardiomyocytes through which mechanism?

- ☐ A. Chelation of sodium ions [2%]
- ☐ B. Allosteric inhibition of OCTN2 [11%]
- ☐ C. Irreversible binding to OCTN2 amino acid residues [2%]
- ☒ D. Competition with L-carnitine for OCTN2 interactions [83%]

Correct

83% answered correctly

Explanation:



Transport proteins **facilitate movement** of specific molecules **across membranes**. A given protein may be able to transport **multiple types of molecules** that have **similar properties** but typically can only interact with **one of these molecules at a time**. Thus, the rate of transport of one molecule may be decreased by the presence of similar molecules, which **compete for interaction** with the transport protein.

Figure 1 shows that L-carnitine and 5-AVAB have *nearly identical* structures, differing only at the hydroxyl (OH) group on L-carnitine and the extra methylene (CH₂) group on 5-AVAB. OCTN2 transports L-carnitine into cells, and 5-AVAB is likely to compete for this interaction due to its similarity to L-carnitine.

The data in Figure 2 show that 5-AVAB decreases the rate of oxygen consumption (likely due to reduced β -oxidation of fatty acids in the mitochondria), and the information in the passage indicates that this is likely due to decreased L-carnitine transport by OCTN2. Figure 2 also shows that increasing the concentration of L-carnitine from 50 μ M to 2 mM **overcomes the effect** of 5-AVAB, indicating that the two molecules **compete** for interaction with OCTN2. Therefore, 5-AVAB most likely alters fatty acid metabolism by **competition with L-carnitine for OCTN2 interactions**.

(Choice A) Chelation is the formation of **coordinate bonds** between a molecule and a metal ion. There is no evidence to indicate that 5-AVAB forms any such bonds with sodium ions.

(Choice B) **Allosteric inhibition** occurs when a molecule binds to a site other than the active site. Because 5-AVAB is similar in structure to L-carnitine, it most likely interacts with OCTN2 at the *same site* as L-carnitine (the active site) and does *not* act allosterically.

(Choice C) If 5-AVAB **irreversibly bound** OCTN2, increasing the L-carnitine concentration to 2 mM would not overcome the effect of 5-AVAB because 5-AVAB would be permanently attached to the protein and would not be displaced by L-carnitine.

Educational objective:

Transport proteins facilitate movement of molecules across membranes. They may transport more than one type of molecule if the molecules have similar properties but can usually interact with only one molecule at a time. Similar molecules may compete for interactions with a transport protein.

Time spent: 0 sec
Subject:
Biochemistry

QID: 402551
System:
Metabolic Reactions

Whole grains have been shown to reduce the risk of cardiovascular disorders. A team of scientists discovered that the compound 5-aminovaleric acid betaine (5-AVAB) was present at high levels in cardiomyocytes of mice given a diet rich in whole grains and decided to investigate its role in metabolism. Previous studies showed that 5-AVAB did not affect glycolysis or citric acid cycle rates in cardiomyocytes when glucose was the primary fuel source, but that 5-AVAB did alter fatty acid metabolism.

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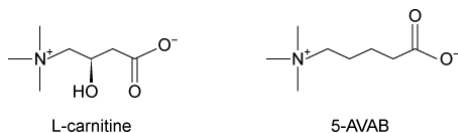


Figure 1 Structures of L-carnitine and 5-AVAB

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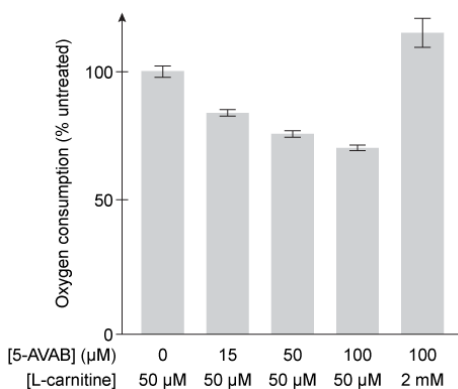


Figure 2 Oxygen consumption rate of cardiomyocytes at various concentrations of 5-AVAB and L-carnitine

In a follow-up experiment, the scientists used siRNA to silence expression of OCTN2 and found that intracellular L-carnitine levels were similarly depleted in siRNA-treated cells as in cells treated with 100 μM 5-AVAB.

OCTN2 facilitates transport of L-carnitine by which mechanism?

- ✓ ☒ A. Active symport [74%]
☐ B. Active antiport [14%]
☐ C. Passive symport [8%]
☐ D. Passive antiport [2%]

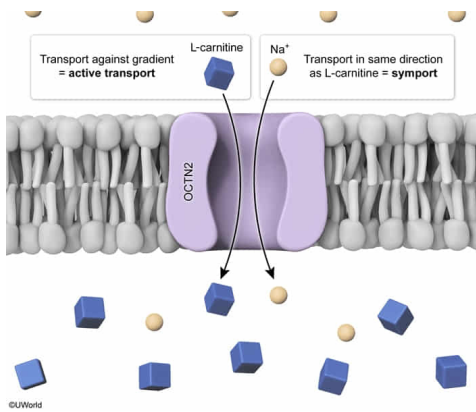
Correct

75% answered correctly

Explanation:

OCTN2 facilitates active symport of L-carnitine





Cell membranes contain multiple types of transport proteins that facilitate the controlled movement of molecules in and out of the cell. The concentration of molecules within the cell is usually different than the concentration of molecules outside the cell, and molecules may travel **down their concentration gradient** (from high to low concentration) or **against their concentration gradient** (from low to high concentration).

Transport across a membrane down a molecule's concentration gradient is referred to as **passive transport** and requires **no energy input**. In contrast, transport against a molecule's gradient **requires energy input** and is called **active transport**. The required energy can be provided by *ATP hydrolysis* or by the simultaneous transport of another molecule *down* its concentration gradient. Transport of a **single molecule** is called **uniport**. When two molecules are transported in the **same direction**, it is known as **symport**, and when two molecules are transported in **opposite directions**, it is called **antiport**.

The passage states that L-carnitine moves into the cell through OCTN2 *against* its concentration gradient, so OCTN2 facilitates **active transport of L-carnitine**. The passage further states that the energy needed for L-carnitine transport is provided by OCTN2 simultaneously transporting Na⁺ into the cell. Because both Na⁺ and L-carnitine are transported in the same direction (into the cell), **OCTN2 facilitates symport**. Therefore, OCTN2 transports L-carnitine into the cell by **active symport**.

(Choice B) Antiport occurs when two molecules are transported through a channel in opposite directions (ie, one into the cell and one out of the cell). Na⁺ and L-carnitine both go into the cell through OCTN2, so OCTN2 facilitates *symport*.

(Choices C and D) Passive transport is the transport of molecules *down* their concentration gradient. L-carnitine is transported *against* its concentration gradient, so it requires active transport.

Educational objective:

Many molecules enter and exit cells through transport proteins. Molecules may travel down their concentration gradient (passive transport, no energy required) or against their concentration gradient (active transport, energy required). Transport proteins facilitate uniport if they transport one molecule, symport if they transport two molecules in the same direction, and antiport if they transport two molecules in opposite directions.

Time spent: 0 sec
Subject:
Biochemistry

QID: 402552
System:
Metabolic Reactions

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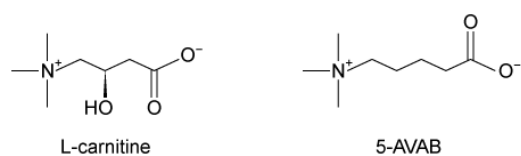


Figure 1 Structures of L-carnitine and 5-AVAB

To test their hypothesis, scientists incubated mouse cardiomyocytes with various concentrations of 5-AVAB for 24 hours. Cells were then provided with the fatty acid palmitate for energy and were deprived of all other energy sources. The cardiomyocytes were also given 50 μM L-carnitine (endogenous extracellular concentration) or 2 mM L-carnitine, and the rate of oxygen consumption was monitored (Figure 2).

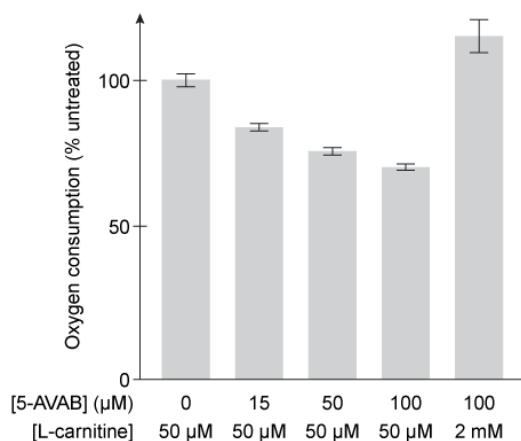
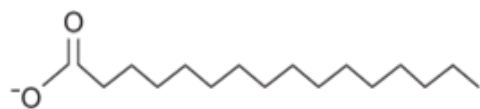


Figure 2 Oxygen consumption rate of cardiomyocytes at various concentrations of 5-AVAB and L-carnitine

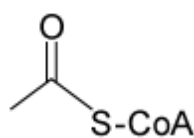
In a follow-up experiment, the scientists used siRNA to silence expression of OCTN2 and found that intracellular L-carnitine levels were similarly depleted in siRNA-treated cells as in cells treated with 100 μM 5-AVAB.

When palmitate is the only energy source, silencing of OCTN2 expression would impair mitochondrial synthesis of which metabolite?



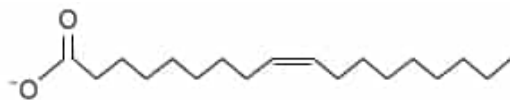
[18%]

✓ ☒ B.



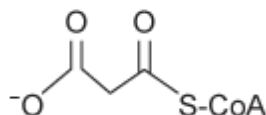
[61%]

☐ C.



[7%]

☐ D.

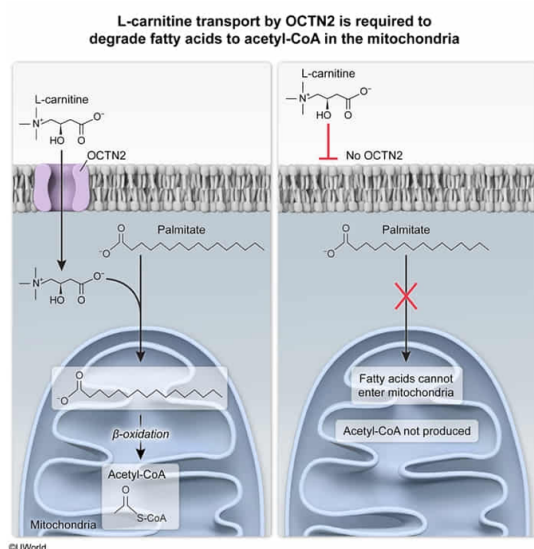


[12%]

Correct

62% answered correctly

Explanation:



The majority of cellular **catabolic reactions** occur in the mitochondria. Many of these catabolic processes **converge** on the molecule **acetyl-CoA**, which can enter the citric acid cycle and provide the reduced electron carriers needed for the **electron transport chain**. **Fatty acids** such as palmitate are **degraded to acetyl-CoA** by **β -oxidation** in the **mitochondria**.

According to the passage, OCTN2 transports L-carnitine into the cell, and this **L-carnitine** is **required for the transport** of **fatty acids** from the cytosol **into the mitochondria**. The passage also states that when OCTN2 expression is silenced by siRNA, intracellular L-carnitine levels are depleted. Therefore, OCTN2 silencing would impair transport of fatty acids into the mitochondria, which would impair their degradation. Because palmitate is the only energy source provided to cells in the experiment, no other source of acetyl-CoA synthesis is available. Therefore, when OCTN2 is silenced, mitochondrial synthesis of **acetyl-CoA** will be impaired.

(Choices A and C) These molecules are examples of fatty acids. Fatty acids are synthesized in

the cytosol, *not* the mitochondria.

(Choice D) This option shows malonyl-CoA, which is used as a precursor in [fatty acid synthesis](#) rather than degradation. All steps of fatty acid synthesis occur in the cytosol, *not* the mitochondria.

Educational objective:

β -oxidation of fatty acids occurs in the mitochondria of eukaryotic cells and produces acetyl-CoA. L-carnitine is required for fatty acids to enter the mitochondria.

Time spent:0 sec

Subject:
Biochemistry

QID:402553

System:
Metabolic Reactions

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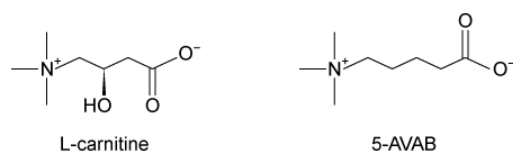


Figure 1 Structures of L-carnitine and 5-AVAB

To test their hypothesis, scientists incubated mouse cardiomyocytes with various concentrations of 5-AVAB for 24 hours. Cells were then provided with the fatty acid palmitate for energy and were deprived of all other energy sources. The cardiomyocytes were also given 50 μM L-carnitine (endogenous extracellular concentration) or 2 mM L-carnitine, and the rate of oxygen consumption was monitored (Figure 2).

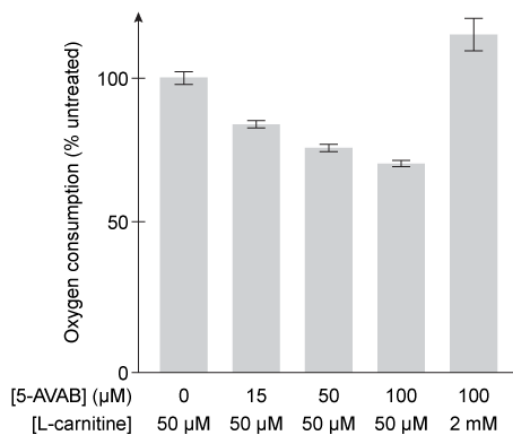


Figure 2 Oxygen consumption rate of cardiomyocytes at various concentrations of 5-AVAB and L-carnitine

In a follow-up experiment, the scientists used siRNA to silence expression of OCTN2 and found that intracellular L-carnitine levels were similarly depleted in siRNA-treated cells as in cells treated with 100 μM 5-AVAB.

The information in the passage and the data in Figure 2 best support which statement about cardiomyocytes when palmitate is their only energy source?

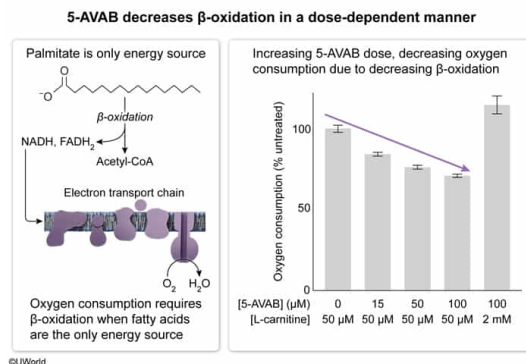
- ☐ A. Cardiomyocytes cannot digest palmitate in the presence of 5-AVAB. [1%]

- ☐ B. 5-AVAB inhibits synthesis of L-carnitine in a dose-dependent manner. [9%]
- ☐ C. Mitochondria cannot import L-carnitine in the presence of 5-AVAB. [7%]
- ✓ ☒ D. 5-AVAB reduces the rate of β -oxidation in a dose-dependent manner. [80%]

Correct

81% answered correctly

Explanation:



The process of **β -oxidation** involves the **degradation of fatty acids** to produce **NADH** and **FADH₂**, which enter the electron transport chain (ETC). Acetyl-CoA is also produced and can enter the citric acid cycle to yield additional NADH and FADH₂. In the ETC, the **electrons from NADH and FADH₂** are ultimately **transferred to oxygen**, which is **reduced to water** in the process.

The passage states that palmitate (a fatty acid) was the only energy source given to cardiomyocytes, and therefore β -oxidation is the only pathway by which these cells can generate NADH and FADH₂. Because β -oxidation is the only pathway that leads to the production of molecules that can enter the ETC in these cells, the rate of oxygen consumption is directly related to the rate of β -oxidation. Figure 2 shows that as the concentration of 5-AVAB *increases*, the rate of oxygen consumption *decreases*. Because palmitate is the only energy source, these data indicate that 5-AVAB **reduces β -oxidation** in a **dose-dependent manner**.

(Choice A) When 5-AVAB is present, oxygen consumption is slowed but not completely halted. Because palmitate is the only energy source provided, these data indicate that cardiomyocytes *can* digest palmitate in the presence of 5-AVAB, but they do so more slowly than in the absence of 5-AVAB.

(Choice B) Figure 2 shows that L-carnitine is *provided* to cardiomyocytes at 50 μ M, or 2 mM. *Synthesis* of L-carnitine is not assessed.

(Choice C) According to the passage, 5-AVAB impedes entry of L-carnitine into the cell from the extracellular space. However, there is no evidence that 5-AVAB inhibits L-carnitine that is already in the cell from entering the mitochondria.

Educational objective:

β -oxidation is the degradation of fatty acids to produce NADH and FADH₂. The electrons from NADH and FADH₂ are used to reduce oxygen to water. Accordingly, oxygen consumption can be used as a measure of electron transport chain activity.

Time spent:0 sec

Subject:
Biochemistry

QID:402554

System:
Metabolic Reactions

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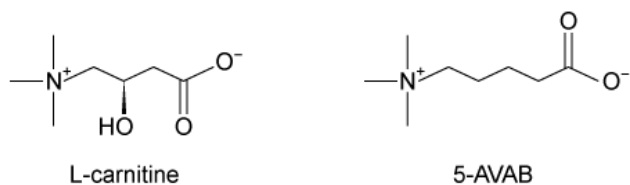


Figure 1 Structures of L-carnitine and 5-AVAB

To test their hypothesis, scientists incubated mouse cardiomyocytes with various concentrations of 5-AVAB for 24 hours. Cells were then provided with the fatty acid palmitate for energy and were deprived of all other energy sources. The cardiomyocytes were also given 50 μM L-carnitine (endogenous extracellular concentration) or 2 mM L-carnitine, and the rate of oxygen consumption was monitored (Figure 2).

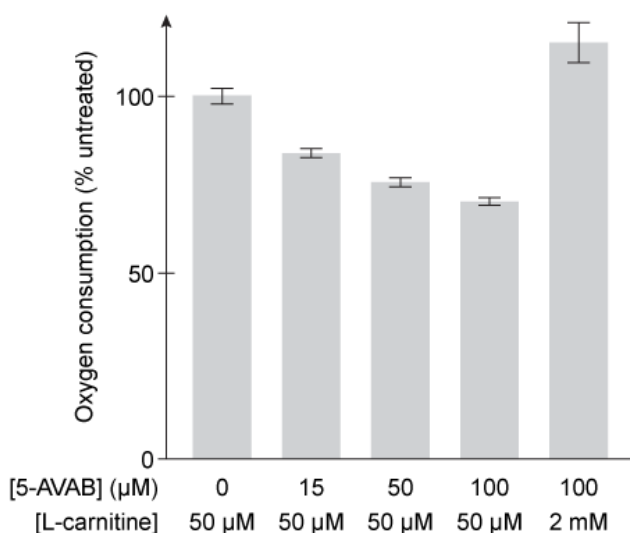


Figure 2 Oxygen consumption rate of cardiomyocytes at various concentrations of 5-AVAB and L-carnitine

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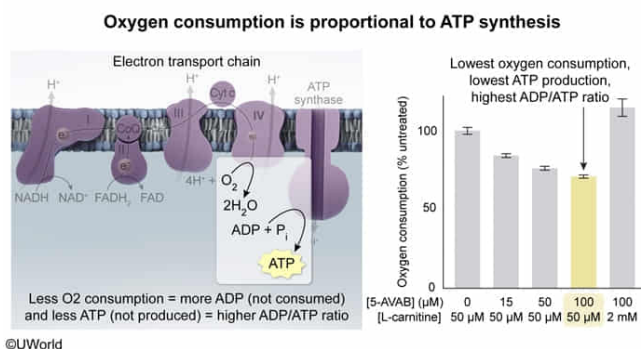
Under which of the conditions tested in Figure 2 would cardiomyocytes have the highest ADP/ATP ratio?

- ☐ A. 0 μM 5-AVAB and 50 μM L-carnitine [3%]
- ☐ B. 15 μM 5-AVAB and 50 μM L-carnitine
- ✓ ☒ C. 100 μM 5-AVAB and 50 μM L-carnitine [69%]
- ☐ D. 100 μM 5-AVAB and 2 mM L-carnitine [26%]

Correct

69% answered correctly

Explanation:



Adenosine triphosphate (ATP) is the main source of energy for cellular processes. In **aerobic organisms**, it is primarily produced in the mitochondria using **energy from the electron transport chain (ETC)** that **drives ATP synthesis** from ADP and inorganic phosphate (P_i). Therefore, **increased ETC activity** corresponds to a **higher level of ATP** and a **lower level of ADP** whereas decreased ETC activity corresponds to the opposite.

The **final step in the ETC** is **reduction of oxygen** to form water, so the **rate of oxygen consumption** serves as a **measure of ETC activity**. The graph in Figure 2 shows that oxygen consumption is lowest in the presence of 100 μM 5-AVAB and 50 μM L-carnitine. Under these conditions ATP will be synthesized

at a slower rate than under any other conditions tested, and ADP will be consumed at a slower rate. Therefore, the **ADP/ATP ratio** will be **highest** in cells at **100 μ M 5-AVAB and 50 μ M L-carnitine**.

(Choices A and B) At 0 μ M or 15 μ M 5-AVAB, in the presence of 50 μ M L-carnitine, the rate of oxygen consumption is *higher* than the rate at 100 μ M 5-AVAB. Accordingly, the ADP/ATP ratio will be *lower* at 0 or 15 μ M 5-AVAB than at 100 μ M 5-AVAB when L-carnitine is held at 50 μ M.

(Choice D) When L-carnitine is present at 2 mM, oxygen consumption occurs at a higher rate than at any other condition tested. ATP will be produced and ADP will be consumed at the fastest rate under these conditions, yielding the highest *ATP/ADP* ratio but the *lowest* ADP/ATP ratio.

Educational objective:

Aerobic organisms primarily generate ATP using energy provided by the electron transport chain (ETC). The final step in the chain is reduction of oxygen to water, and the rate of oxygen consumption can be used as a measure of ETC activity. Higher ETC activity corresponds to a higher ATP/ADP ratio or a lower ADP/ATP ratio.

Time spent:0 sec

Subject:
Biochemistry

QID:402555

System:
Metabolic Reactions

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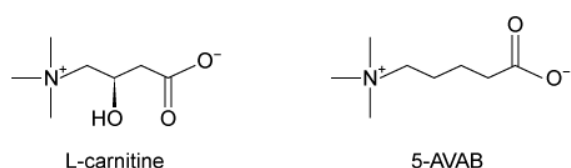


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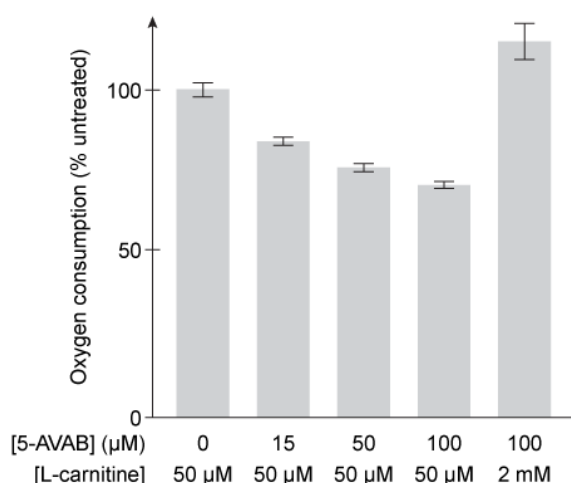


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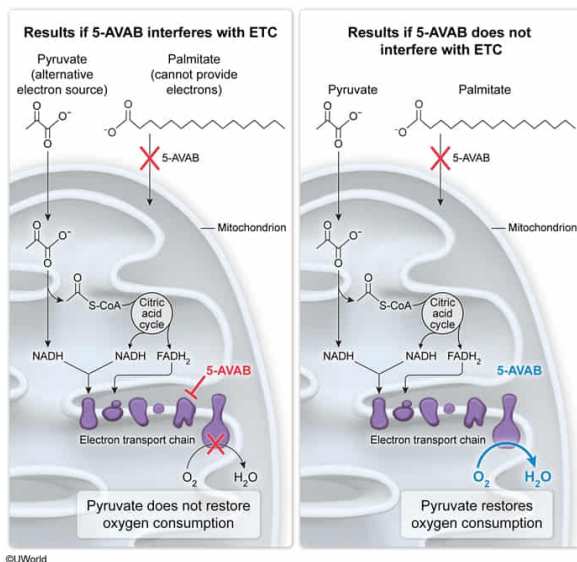
A colleague suggested that 5-AVAB may also interfere with the electron transport chain directly. Based on the information in the passage, which experiment could provide evidence for the effect (if any) of 5-AVAB on the electron transport chain?

- ✔ ☒ A. Incubate cardiomyocytes with pyruvate and measure oxygen consumption in the presence and absence of 5-AVAB. [38%]
- ☐ B. Add an electron transport chain inhibitor to each set of cardiomyocytes and measure the change in oxygen consumption in the presence and absence of 5-AVAB. [37%]
- ☐ C. Increase the concentration of palmitate given to cardiomyocytes and measure oxygen consumption in the presence and absence of 5-AVAB. [8%]
- ☐ D. Increase the partial pressure of oxygen to which the cardiomyocytes are exposed and measure the rate of oxygen consumption in the presence and absence of 5-AVAB. [15%]

Correct

38% answered correctly

Explanation:



The **electron transport chain (ETC)** uses **electrons from NADH and $FADH_2$** to reduce oxygen to water, providing the energy needed for ATP synthesis. NADH and $FADH_2$ are largely produced in the **citric acid cycle** and **several metabolic pathways lead to this cycle**, including **glycolysis** and **β -oxidation**.

The scientists hypothesized that 5-AVAB inhibits fatty acid degradation (β -oxidation) in the mitochondria and thereby decreases oxygen consumption. They further hypothesized that this inhibition is accomplished by blocking L-carnitine from entering cells, impairing L-carnitine-mediated transport of fatty acids into the mitochondria. Although the data in Figure 2 are consistent with this hypothesis, it is possible that 5-

AVAB decreases oxygen consumption by directly inhibiting the ETC. To test this possibility, an **alternative source of NADH and FADH₂** must be provided. Pyruvate produces NADH by conversion to acetyl-CoA. Acetyl-CoA then enters the citric acid cycle.

The passage states that **5-AVAB does not affect glycolysis or citric acid cycle rates**, so **addition of pyruvate allows for direct testing of the effect of 5-AVAB on the ETC**. If 5-AVAB inhibits the ETC directly, then cells provided with pyruvate will have a higher oxygen consumption rate in the *absence* of 5-AVAB than in its *presence*. Conversely, if 5-AVAB does *not* directly impact the ETC, then cells provided with pyruvate will have the *same* oxygen consumption rate *regardless* of whether 5-AVAB is present.

(Choice B) Adding a known ETC inhibitor will reduce oxygen consumption in *all* cells, whether or not 5-AVAB is present. The presence of such an inhibitor could confound any effect that 5-AVAB might have and would not reveal whether any effects of 5-AVAB are due to direct interference with the ETC or interference upstream (eg, L-carnitine transport).

(Choices C and D) Palmitate and oxygen are consumed by β -oxidation and the ETC, respectively. Altering their concentration or partial pressure will not give any information about the effect of 5-AVAB on ETC activity.

Educational objective:

The electron carriers needed for the electron transport chain (ETC), NADH and FADH₂, can be provided by several pathways including glycolysis, β -oxidation, and the citric acid cycle. A molecule that affects the ETC directly will block the use of electron carriers regardless of which pathway generates them.

Time spent:0 sec
Subject:
Biochemistry

QID:402556
System:
Metabolic Reactions